

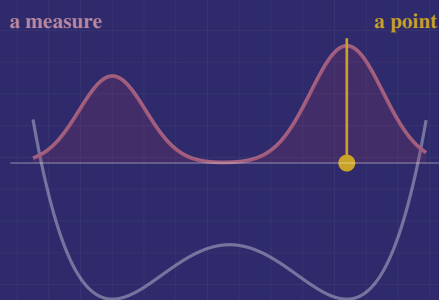
what mathematics molecular biology is actually missing

The Measure, Not the Point

Mutation, docking and structure prediction are the same unsolved problem

Three biological questions. One mathematical obstruction.

A three-tier honest sort of what is engineering, what is translation, and what does not yet exist.



TIER 1

Engineering

existing statistics, applied correctly

TIER 2

Translation

deep maths that exists, not yet carried across

TIER 3

Genuinely new

mathematics that does not exist yet

The thesis. In every one of these three problems we have a good theory of the *individual object* — a structure, a pose, a base pair — and **no tractable theory of the measure over those objects, nor of the operator that evolves that measure in time.**

The obstruction has a name and a complexity class. It is the partition function $Z = \int e^{-\beta U(x)} dx$, and evaluating it is **#P-hard**. So we substitute point estimates for measures everywhere — and then find that biology, which is entirely about ensembles and rates, does not follow.

What is in this dossier

- the three problems shown to be one
- what AlphaFold 3 did *not* crack
- the live attack on Z
- proton tunnelling in DNA, precisely
- the transfer operator, made concrete
- a tier scoreboard, and where to bet

1	Three problems, one shape	2
1.1	Why this is not a complaint that “biology is complicated”	2
2	The villain: why everything reduces to Z	4
2.1	What we do instead — and what it costs us	4
3	“Apparent random” mutation — the word <i>apparent</i> is doing real work	6
3.1	Level 1 — not uniform in space or time	6
3.2	Level 2 — the quantum substrate	6
4	Docking and dynamics: the error is entropic	8
4.1	A small difference of two enormous numbers	8
4.2	The two entropy terms that dominate, and neither is solved	8
4.3	The other wall: fifteen orders of magnitude	8
4.4	Where genuinely new mathematics has already landed	9
5	AlphaFold 3: exactly what it cracked, and what it left untouched	10
5.1	What AF3 is, mathematically	10
5.2	The honest ledger	10
6	The missing object, made concrete: the transfer operator	11
7	The live attack on Z	12
7.1	Boltzmann generators	12
7.2	The wider family this belongs to	12
8	The scoreboard	13
9	Where I would bet	13
10	Source ledger	15
10.1	How this dossier relates to the others	15

The three-tier sort used throughout

TIER 1 **Engineering** — existing statistics applied correctly. Part VIII was this, and said so.

TIER 2 **Translation** — deep mathematics that already exists and has not been carried into biology. Cheap, fast, underexploited.

TIER 3 **Genuinely new** — mathematics that does not exist. Rare, slow, and the only place a mathematician is strictly necessary.

Knowing which tier a problem sits in is what stops a student spending four years on something that was either already solved or provably impossible.

1 Three problems, one shape

You named three things: apparently random mutation, realistic docking and dynamics, and what AlphaFold 3 did or did not solve. They look like three fields. They are one mathematical failure, and the failure has a precise form.

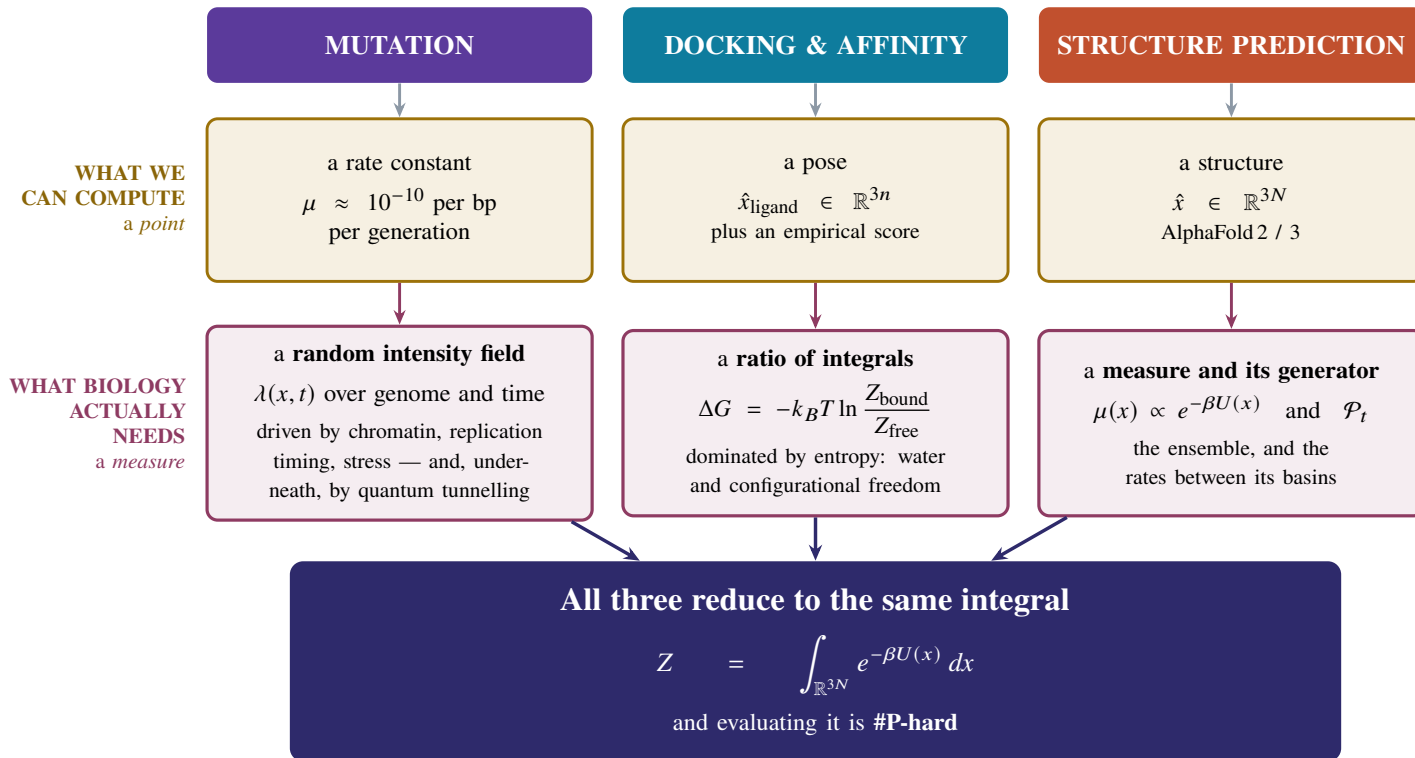


Figure 1. The centrepiece of this dossier. Read it downward. The top row is what each field delivers today; the second row is what the biology actually is; the bottom is why the second row is hard. **Nothing in the middle row is a modelling preference. Each is the literal mathematical object the biology consists of.**

The thesis, stated once

*In every case we have a good theory of the individual object,
and no tractable theory of the measure over those objects,
or of the operator that evolves that measure in time.*

A structure predictor returns a point. Biology needs $(\mu, \mathcal{P}_t)$. A docking program returns a pose. Biology needs ΔG , which is a ratio of integrals over μ . A mutation model returns a constant. Biology needs a stochastic field. **The gap is the same gap three times.**

1.1 Why this is not a complaint that “biology is complicated”

It is a specific, nameable, provable computational obstruction with a specific mathematical consequence, and that distinction matters — because obstructions can be attacked and complexity cannot.

The vague version (not useful)	The precise version (attackable)
“Cells are too complex to model.”	Z is #P-hard, so we substitute $\arg \max_x e^{-\beta U(x)}$ for $\int e^{-\beta U(x)} dx$ — a <i>mode</i> for a <i>measure</i> .
“Docking scoring functions are inaccurate.”	ΔG is a small difference of large numbers whose dominant terms are entropic, and entropy <i>is</i> $\ln Z$.
“AlphaFold doesn’t capture dynamics.”	It estimates a mode of μ . It does not estimate μ , and it does not estimate the transfer operator $\mathcal{P}_t$ that governs transitions between the modes.
“Mutations are random.”	They are a Cox process — a Poisson process whose intensity is itself a random field — with a partly quantum-mechanical substrate.

2 The villain: why everything reduces to Z

Everything a molecule does at equilibrium follows from one number.

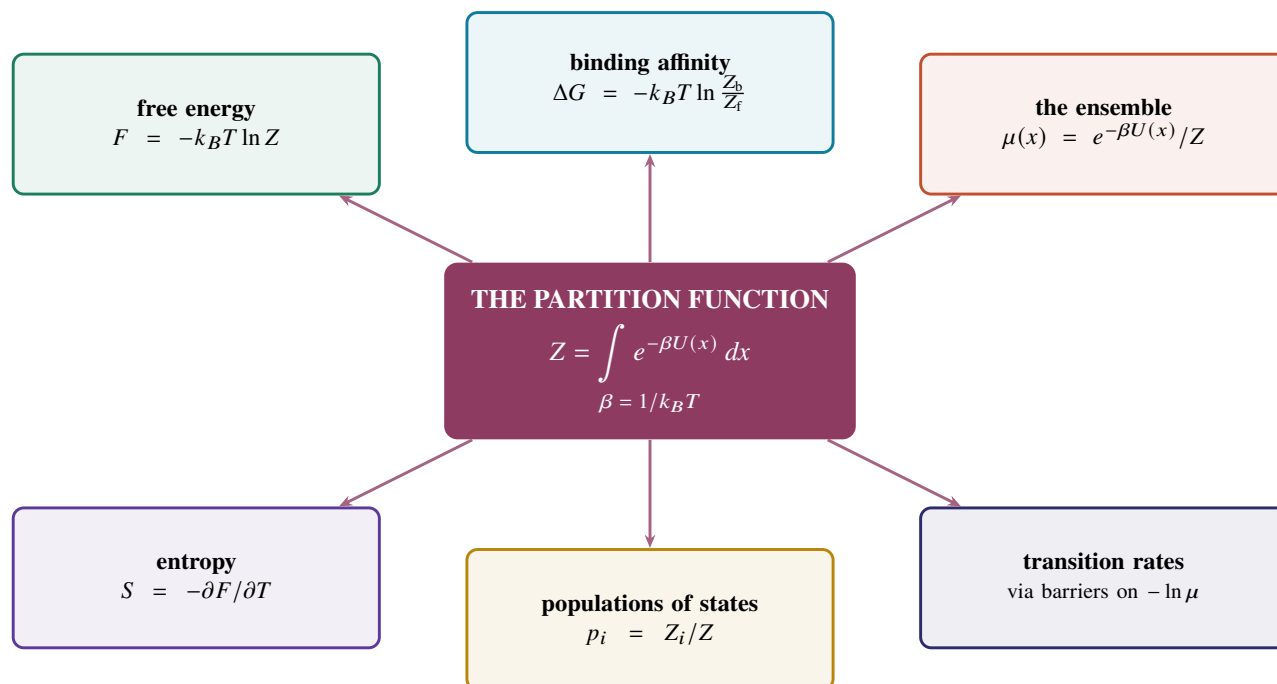


Figure 2. Six quantities molecular biology asks for daily. All six are functions of one integral — and we cannot compute it.

Why it is genuinely hard, not merely expensive

Z is a $3N$ -dimensional integral over a function with an enormous number of sharp, well-separated maxima. For a modest protein $N \sim 10^4$ atoms, so the integral is over $\mathbb{R}^{30,000}$.

No quadrature scheme reaches that dimension. The only general approach is Monte Carlo — and Monte Carlo converges at $O(n^{-1/2})$ *provided it mixes*. On a rugged landscape with barriers $\gg k_B T$, it does not mix: escape from a basin takes time $\propto e^{\beta \Delta U}$, so sampling time is *exponential in the barrier height*.

Approach	What it costs	Where it breaks
brute-force MD	10^{-15} s timesteps	needs 10^{17} steps to reach 100 s
free energy perturbation (FEP)	GPU-days per ligand <i>pair</i>	only between chemically similar ligands
metadynamics / umbrella sampling	hours–days	requires guessing good collective variables
exact evaluation	—	#P-hard

2.1 What we do instead — and what it costs us

Because Z is out of reach, every practical method quietly replaces the measure with its mode:

$$\underbrace{\mu(x) = \frac{e^{-\beta U(x)}}{Z}}_{\text{what the molecule is}} \longrightarrow \underbrace{\hat{x} = \arg \max_x e^{-\beta U(x)}}_{\text{what we compute}}$$

This substitution is invisible in the output — you get a structure, or a pose, or a number — and it is exactly wrong whenever the measure is **multimodal**, which for a functioning protein it always is.

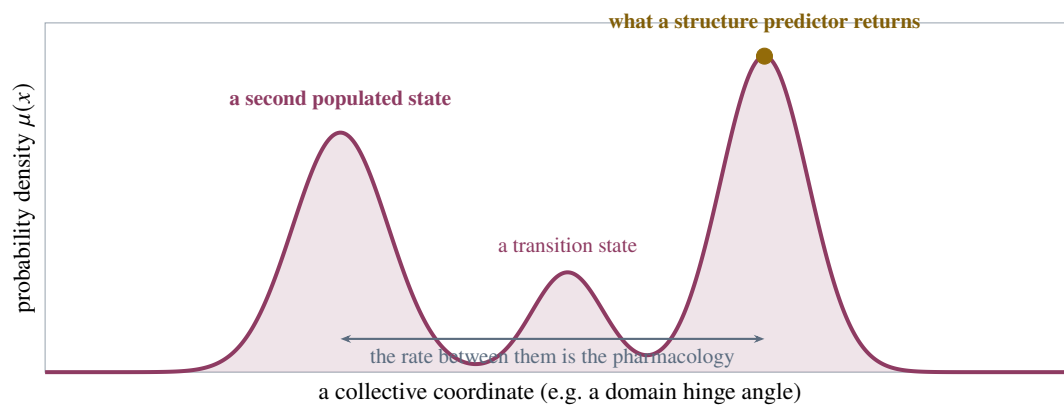


Figure 3. A schematic conformational measure. The gold point is the entire output of a structure prediction. Everything that makes the protein a *machine* — the second state, the barrier between them, the residence time — is in the part that is discarded.

3 “Apparent random” mutation — the word *apparent* is doing real work

Your phrasing was exact. Mutation is *apparently* random on two independent levels, and they need separating because they sit in different tiers.

3.1 Level 1 — not uniform in space or time

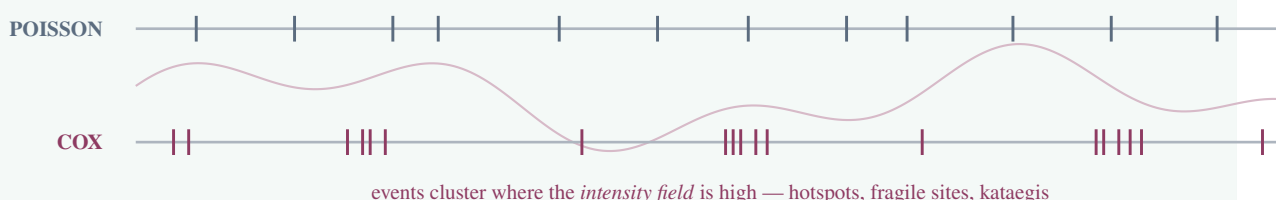
TIER 2 *established; the mathematics exists and is simply not used here.*

Finding	Evidence
Mismatch repair efficiency tracks chromatin state, so mutation rate varies more than two-fold across the genome	Supek & Lehner, <i>Nature</i> 521 , 81 (2015)
~80+ distinct mutational signatures decompose the 96-channel substitution spectrum	Alexandrov et al., <i>Nature</i> 500 (2013); PCAWG, <i>Nature</i> 578 (2020)
CpG deamination produces ~10-fold local hotspots	Lindahl, <i>Nature</i> 362 , 709 (1993)
Mutation rate is <i>regulated</i> : stress induces error-prone polymerases (SOS response)	Rosenberg and successors — mutation rate is a controlled variable, not a constant
Mutation bias correlates with gene essentiality in <i>Arabidopsis</i>	Monroe et al., <i>Nature</i> 602 (2022) — contested ; treat as unsettled

The wrong object, and the right one

Standard practice models mutation as a **homogeneous Poisson process** with rate μ . If the rate varies across the genome and over time, the correct object is a **Cox process** — a Poisson process whose intensity is itself a random field:

$$\lambda(x, t) = \lambda_0 \exp(f(\text{chromatin, replication timing, transcription, stress}))$$



Why this is not academic, for your AMR work: if the intensity is a field rather than a constant, the expected waiting time to a resistance mutation is *not* $1/\mu$. Clustered intensity makes rare-but-fast emergence far more likely than a Poisson model predicts. That changes stewardship and dosing predictions, and the mathematics (log-Gaussian Cox processes, Hawkes processes for clustered events) is completely standard and completely unused here.

3.2 Level 2 — the quantum substrate

TIER 3 *this is where the mathematics genuinely runs out.*

This is where your question actually points, and it has a precise, citable literature.

Löwdin’s proposal, and its modern treatment

Löwdin (1963), *Rev. Mod. Phys.* **35**, 724, proposed that point mutations arise from **proton tunnelling** across the hydrogen bonds of a base pair, producing rare **tautomers** (G–C, A–T) that mispair when the replication fork arrives.

Slocombe, Sacchi & Al-Khalili, *Communications Physics* **5**, 109 (2022) — “An open quantum systems approach to proton tunnelling in DNA” — treated the G–C proton transfer as a double well coupled to a thermal bath and found the tautomeric population reaches $\approx 1.7 \times 10^{-4}$, equilibrating on **sub-picosecond** timescales: orders of magnitude faster and more populated than a classical over-barrier estimate.

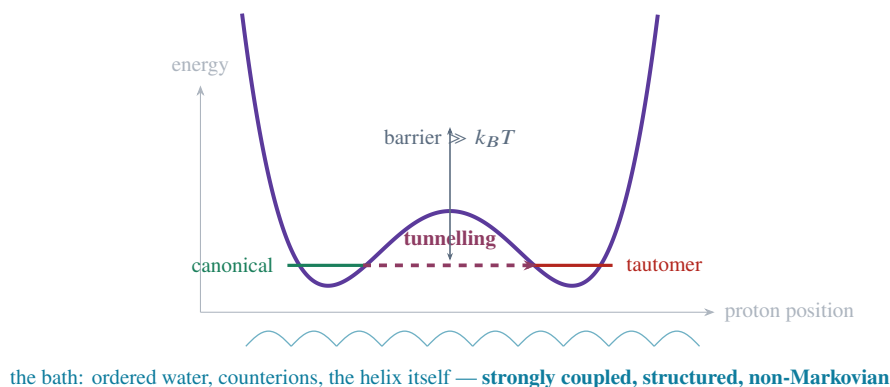


Figure 4. The Löwdin picture as an open quantum system. Everything hard is in the bath: it is not weak, not memoryless, and not generic.

What mathematics is actually missing here

Missing object	Why it is hard	State of the art
Non-Markovian dynamics at <i>strong</i> coupling	Lindblad needs weak coupling <i>and</i> memorylessness — both violated. HEOM is numerically exact but scales exponentially in bath complexity	Genuinely open. No tractable general theory
Decoherence in <i>structured</i> environments	Tegmark-style estimates assume a generic bath. DNA's environment is ordered and correlated	No theory
Quantum-origin error meeting a <i>classical</i> proofreader	Kinetic proofreading (Hopfield 1974) is entirely classical. There is no theory of a classical corrector acting on a quantum-origin error	Not even properly posed

Be careful here — and tell your students the same

The tunnelling is real physics. **The link to mutation is not closed.** The tautomer must survive until the replication fork arrives *and* the polymerase must misread it, and tautomer lifetimes are short. Most spontaneous mutation is attributed to replication error, deamination and oxidative damage, not tunnelling. This is the same epistemic situation as the photosynthesis coherence claims in Part VII: genuine quantum mechanics, unproven functional consequence. **Mark it as open, work on it if you want to do mathematics, and do not cite it as established.**

4 Docking and dynamics: the error is entropic

Docking's weakness is not geometry. Poses are usually roughly right. The failure is in the number attached to the pose, and the reason is arithmetic before it is chemistry.

4.1 A small difference of two enormous numbers

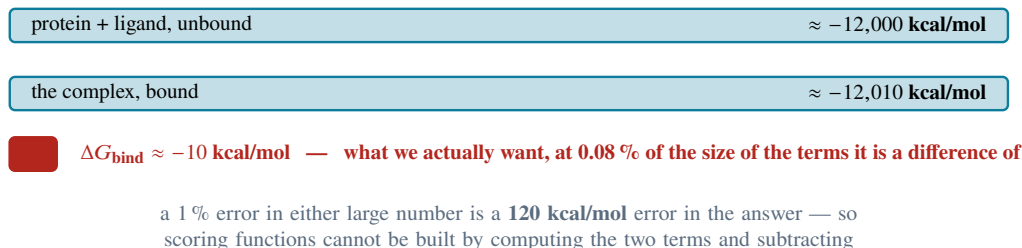


Figure 5. Illustrative magnitudes, to scale. This is why empirical scoring functions are fitted regressions rather than physics: the physics route requires cancellation to five significant figures.

And the part that will not cancel is entropic:

$$\Delta G_{\text{bind}} = \underbrace{\Delta H}_{\text{computable-ish}} - T \underbrace{\Delta S}_{\text{the problem}} \quad \text{and} \quad S = -\frac{\partial F}{\partial T}, \quad F = -k_B T \ln Z.$$

Entropy is $\ln Z$. So docking's central difficulty is the same villain as §2.

4.2 The two entropy terms that dominate, and neither is solved

Term	Why it dominates	Best available, and its limit
Desolvation (ordered water)	Displacing structured water from a binding site can supply most of the affinity. The water is <i>part of the receptor</i> , and it is invisible in most crystal structures.	Inhomogeneous solvation theory — WaterMap, Abel et al., <i>JACS</i> 130 , 2817 (2008). No rigorous tractable theory of which waters are displaceable, and at what cost.
Configurational entropy	Both partners lose conformational freedom on binding. This is an integral over a $3N$ -dimensional manifold.	Free energy perturbation: ~ 1 kcal/mol accuracy demonstrated (Wang et al., <i>JACS</i> 137 , 2695, 2015) — but GPU-days per ligand pair , and only between chemically similar ligands, because the alchemical path must be smooth.

4.3 The other wall: fifteen orders of magnitude

Binding *kinetics* — residence time $1/k_{\text{off}}$ — predicts clinical efficacy better than affinity does for many targets. It is also far harder, because it is a rare-event problem.

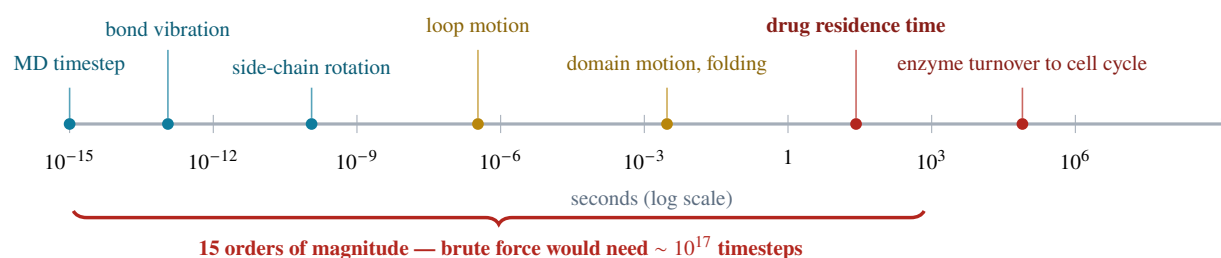


Figure 6. The sampling gap. Enhanced sampling closes part of it by biasing the measure; it does not close it by brute force, and it never will.

4.4 Where genuinely new mathematics has already landed

This is the encouraging part of your question: the answer is that it *is* happening.

Advance	The mathematics	Why it matters
E(3)-equivariant networks MACE (Batatia et al., NeurIPS 2022); NequIP; Allegro	Equivariant message passing built on higher-order tensor products of irreducible representations of $E(3)$ (Thomas et al. 2018; e3nn)	DFT-level accuracy at near-classical cost. Representation theory doing load-bearing work in a force field. This one genuinely worked.
Boltzmann generators Noé et al., <i>Science</i> 365 , eaaw1147 (2019)	Normalizing flows: learn an invertible f from a Gaussian to the Boltzmann measure, with tractable Jacobian, then reweight to exactness	A direct assault on Z. Sample the measure without running the trajectory. See §7.
Learned collective variables VAMPnets (Mardt et al., <i>Nat Commun</i> 9 , 5, 2018)	Koopman / transfer-operator spectral theory (Wu & Noé, <i>J Nonlinear Sci</i> 30 , 2020)	Finds the slow coordinates instead of guessing them. See §6.
Metadynamics Laio & Parrinello, <i>PNAS</i> 99 , 12562 (2002)	Adaptive history-dependent bias on a free-energy surface	Mature and widely used; the CV-choice problem is precisely what VAMPnets repair.

5 AlphaFold 3: exactly what it cracked, and what it left untouched

I do not want to undersell this. AF2 and AF3 are a genuine landmark. But the precise statement of what was solved is what tells you where to work next.

5.1 What AF3 is, mathematically

Abramson et al., *Nature* 630, 493 (May 2024)

AF2's structure module used **Invariant Point Attention** — SE(3)-equivariant attention over per-residue frames (a rotation and a translation per residue), so the prediction is independent of how you orient the whole molecule.

AF3 replaced that module with a diffusion model operating directly on atomic coordinates, and extended coverage to nucleic acids, ligands, ions and modified residues. Equivariance is traded for a generative denoiser plus data augmentation.

Note the irony, and it is a useful one: **AF3's structure module is already a generative model of a distribution** — it just gets collapsed to a single ranked output. The machinery for producing a measure is partly there; what is missing is that the model is not trained to make that measure the *Boltzmann* measure.

5.2 The honest ledger

Left untouched	Why it matters	State of play
Conformational ensembles	Function lives in the ensemble: allostery, cryptic pockets, induced fit	MSA-subsampling hacks (Del Alamo et al., <i>eLife</i> 11, e75751, 2022); AlphaFlow and ensemble emulators are early and unvalidated
Free energies and affinities	AF3 returns a <i>pose</i> , not ΔG	Affinity prediction remains the open problem; see §4
Effect of a point mutation	A single substitution barely changes the MSA, so the prediction barely moves	AF2 performs poorly on $\Delta\Delta G$ of stability — repeatedly documented
Intrinsic disorder	~30 % of the human proteome	Low-pLDDT spaghetti: honest, but not usable
Kinetics	Folding paths, transition states, residence time — the actual pharmacological variable	Entirely outside the model
Novel folds, shallow MSAs	Performance degrades where there is no family to lean on	The same failure mode flagged for AMR in Part VII: a learned prior regresses the novel toward the common

The one-sentence statement

*AlphaFold solved a structure problem.
Biology needs an ensemble-and-rate problem solved.*

These are different mathematical objects. Formally, what is missing is the Boltzmann measure μ and the **transfer operator** $\mathcal{P}_t$ that propagates it. A structure predictor returns a point. Biology needs $(\mu, \mathcal{P}_t)$ — and the second one has barely been started.

IN PLAIN WORDS

AlphaFold gave us a very good photograph of a machine. What we need is a film of it working, and a measurement of how fast each part moves. A photograph is not a bad film — it is a different kind of object, and no amount of sharpening turns one into the other.

6 The missing object, made concrete: the transfer operator

TIER 2 This mathematics is mature. It is simply not used in pharmacology.

A structure predictor gives you a point in conformation space. The object that propagates a *distribution* over that space has a name.

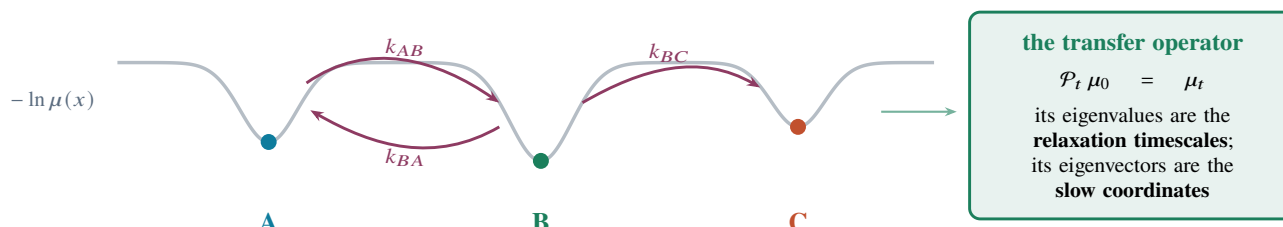


Figure 7. A structure predictor returns basin B (the deepest). The pharmacology is in k_{AB} , k_{BA} , k_{BC} — and in whether basin A exists at all, which is where a cryptic pocket would be.

Why this is the right object, and not just a nice picture

Let μ_0 be any starting distribution over conformations. The **transfer operator** $\mathcal{P}_t$ (Perron–Frobenius) propagates it forward by time t ; its adjoint is the **Koopman operator**, which propagates observables instead of densities. Three facts make this the natural object:

1. Its **leading eigenvector is the equilibrium measure** μ . So the ensemble problem and the kinetics problem are the *same* spectral problem.
2. Its **remaining eigenvalues give the relaxation timescales** $t_i = -\tau/\ln \lambda_i$ — exactly the rates you want, with no need to define “the” reaction coordinate first.
3. The **eigenvectors are the slow collective variables**. The CV-choice problem that plagues metadynamics is solved rather than guessed.

And it is learnable from short simulations. You do not need one trajectory that crosses every barrier; you need many short ones that each sample locally, and the operator stitches them together. This is what Markov state models do, and what VAMPnets (Mardt et al., *Nat Commun* **9**, 5, 2018) learn directly, on the theoretical footing of VAMP (Wu & Noé, *J Nonlinear Sci* **30**, 2020).

The gap worth pointing a student at

Koopman theory is mature. VAMPnets showed it can be learned from data. **Almost nobody in drug discovery uses it to predict k_{off} .** Predicting residence time rather than affinity would change medicinal chemistry more than a better docking score would — and unlike the Tier 3 problems, the mathematics already exists.

7 The live attack on Z

If Z is the villain, the interesting question is who is attacking it. One idea stands out, because it does not try to sample the measure faster — it tries to *learn the measure directly*.

7.1 Boltzmann generators

Noé, Olsson, Köhler & Wu, *Science* 365, eaaw1147 (2019)

Instead of running molecular dynamics until it mixes, **learn an invertible map f** from a simple Gaussian in latent space to configuration space, trained so that pushing the Gaussian through f approximates the Boltzmann distribution. Because f is invertible with a tractable Jacobian, you know the density of every generated sample exactly — so you can **reweight to the true Boltzmann distribution** and recover unbiased equilibrium statistics.

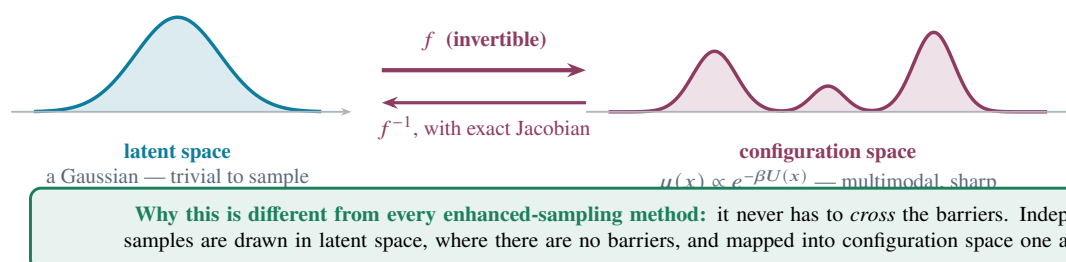


Figure 8. The Boltzmann generator idea. This is the most genuinely new piece of mathematics in the whole of this dossier: it changes the question from “how do I sample this measure” to “can I learn a change of variables that makes it easy”.

The honest caveat — and it is a serious one

Boltzmann generators have been demonstrated on **small systems**. Whether they scale to a protein–ligand complex is **open**: the flow must learn a measure in $\mathbb{R}^{3N}$ with $N \sim 10^4$, sharp, multimodal, and with topology that changes as bonds rotate.

That is precisely the right situation for a mathematician. It is a clearly posed problem, with a working prototype, a known obstruction, and no known solution. Compare that with “model the cell”, which is none of those things.

7.2 The wider family this belongs to

Idea	What it learns	Status
Normalizing flows	an invertible change of variables with exact density	works; scaling open
Diffusion models	a denoising process whose stationary law is the target	what AF3’s structure module already is — but not trained to the <i>Boltzmann</i> target
Flow matching (Lipman et al. 2023)	a velocity field transporting one measure to another	cheaper training; being applied to ensembles now
Optimal transport	the geometry of the space of measures itself	the natural language for comparing two ensembles, and underused

The unifying observation

Every entry in that table is an answer to the same question: **how do you represent and manipulate a probability measure in very high dimension?** That is a mathematical question, not a biological one — which is exactly why it is tractable, and exactly why biology should be paying attention to it.

8 The scoreboard

The honest sort. This is the table to argue with.

Problem	Tier	What is actually blocking it
Mutation as a non-uniform random field	TIER 2	Cox and Hawkes processes exist and are standard. Nobody is using them in AMR emergence modelling.
Proton tunnelling → mutation	TIER 3	Non-Markovian dynamics at strong coupling in a structured bath. No tractable theory exists.
Docking: water and desolvation	TIER 3	No rigorous tractable theory of which waters are displaceable, and at what free-energy cost.
Docking: configurational entropy	TIER 3	Z is #P-hard. Boltzmann generators are the live attack; scaling unsolved.
Force-field accuracy	solved-ish	E(3)-equivariant ML potentials (MACE, NequIP) reached DFT accuracy at classical cost. This one genuinely worked.
Protein conformational ensembles	TIER 2 → TIER 3	Koopman/VAMP theory exists; generative ensemble models are new and largely unvalidated.
Binding kinetics, k_{off}	TIER 3	Rare events spanning 15 orders of magnitude in time.
Long-timescale MD	not maths	Hardware. Do not send a mathematician.
Multiscale coupling QM → MM → continuum	TIER 3	Mori–Zwanzig gives an exact but intractable memory kernel. Learning it is open.
Paralogue vs orthologue calling [PART VIII §7]	TIER 1	Likelihood ratios and profile HMMs. Engineering.
Structured instrument error [PART VIII]	TIER 1	Robust statistics. A 50-line script.

Table 1. **TIER 1** existing statistics applied correctly across **TIER 3** mathematics that does not yet exist. **TIER 2** deep mathematics that exists but has not been carried

9 Where I would bet

Three directions, ranked by how likely I think each is to produce something genuinely new within a decade.

1 · Generative samplers of Boltzmann measures

Normalizing flows, diffusion and flow matching used not to *predict a structure* but to *sample a distribution* with correct statistical weights.

Why: it attacks Z, the common villain in all three of your problems.

Horizon: usable output in ~2 years.

2 · Transfer-operator learning for kinetics

Koopman theory is mature; VAMPnets showed it can be learned from short simulations; almost nobody in pharmacology uses it.

Why: predicting k_{off} rather than ΔG would change drug discovery more than better docking would.

Horizon: usable output in ~2 years.

3 · Open quantum systems in warm structured environments

Proton tunnelling in DNA, radical-pair magnetoreception and enzymatic hydrogen tunnelling are the *same* mathematical problem in three biological costumes.

Why: it is the only one where the mathematics genuinely does not exist.

Horizon: decade-scale. Highest risk, highest novelty.

IN PLAIN WORDS

If a student wants to *do mathematics*, send them to number 3. If a student wants to *change what the lab can measure within a PhD*, send them to number 1 or 2. Both are honourable; confusing them wastes years.

And the caution I keep repeating, because it keeps being true

For most of what a molecular biology lab does day to day, the binding constraint is still **measurement**, not mathematics. Part VIII made that case with your own sequencing data: half the errors came from four moments on a flow cell, and no new mathematics was required to find them.

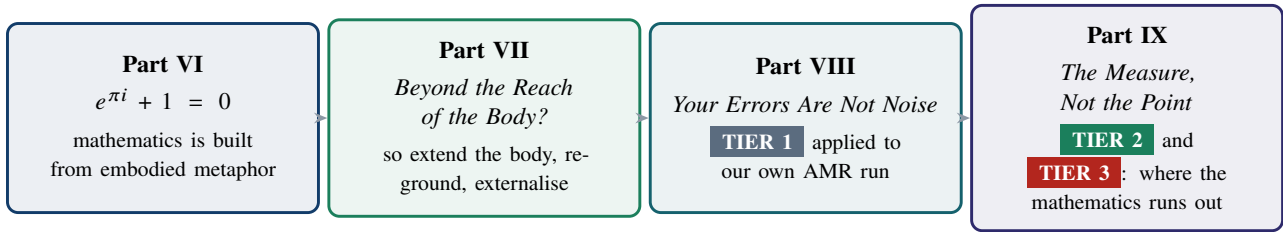
New mathematics is the right answer to a specific, well-posed, blocked calculation. It is almost never the right answer to “our results are noisy”.

10 Source ledger

Every substantive claim, with its author and year. Contested items are marked.

Year	Result	Who / where, and status
Mutation		
1963	Proton tunnelling produces mutagenic tautomers	Löwdin, <i>Rev. Mod. Phys.</i> 35 , 724
1974–75	Kinetic proofreading (classical)	Hopfield, <i>PNAS</i> 1974; Ninio 1975
1993	Spontaneous DNA decay; deamination rates	Lindahl, <i>Nature</i> 362 , 709
2013–20	Mutational signatures from the 96-channel spectrum	Alexandrov et al., <i>Nature</i> 500 ; PCAWG, <i>Nature</i> 578
2015	Mismatch repair tracks chromatin $\Rightarrow$ >2-fold rate variation	Supek & Lehner, <i>Nature</i> 521 , 81
2022	Mutation bias correlates with essentiality in <i>Arabidopsis</i>	Monroe et al., <i>Nature</i> 602 CONTESTED
2022	Open-quantum-systems treatment of G–C proton transfer; tautomer population $\approx 1.7 \times 10^{-4}$	Slocombe, Sacchi & Al-Khalili, <i>Commun. Phys.</i> 5 , 109 LINK TO MUTATION NOT CLOSED
Docking, dynamics, sampling		
2002	Metadynamics	Laio & Parrinello, <i>PNAS</i> 99 , 12562
2008	Inhomogeneous solvation theory (WaterMap)	Abel et al., <i>JACS</i> 130 , 2817
2015	FEP+ reaching ~ 1 kcal/mol on congeneric series	Wang et al., <i>JACS</i> 137 , 2695
2018	VAMPnets: learning the transfer operator from data	Mardt, Pasquali, Wu & Noé, <i>Nat Commun</i> 9 , 5
2018	Tensor field networks; e3nn	Thomas et al.; Geiger & Smidt
2019	Boltzmann generators	Noé, Olsson, Köhler & Wu, <i>Science</i> 365 , eaaw1147
2020	VAMP variational theory	Wu & Noé, <i>J Nonlinear Sci</i> 30
2022	MACE: equivariant higher-order message passing	Batatia et al., NeurIPS
2023	Flow matching	Lipman et al.
Structure prediction		
2021	AlphaFold 2; Invariant Point Attention	Jumper et al., <i>Nature</i> 596
2022	Conformational ensembles via MSA subsampling	Del Alamo et al., <i>eLife</i> 11 , e75751
2023	ESMFold / ESM-2 protein language model	Lin et al., <i>Science</i> 379
2024	AlphaFold 3 : diffusion over atomic coordinates	Abramson et al., <i>Nature</i> 630 , 493
Mathematical background		
—	Partition function evaluation is #P-hard	standard complexity result for the general case
—	Perron–Frobenius / Koopman operator duality	classical ergodic theory
—	Mori–Zwanzig projection and the memory kernel	classical non-equilibrium statistical mechanics
1997–99	Non-equilibrium work relations	Jarzynski, <i>PRL</i> 78 ; Crooks, <i>PRE</i> 60

10.1 How this dossier relates to the others



We can compute points but not measures, states but not rates.

Mutation, docking and folding are three faces of that one sentence. The obstruction is the partition function, and the most promising attack on it is to stop sampling the measure and start learning it.